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# Protective effect of HwangRyunHaeDok-Tang water extract against chronic obstructive pulmonary disease induced by cigarette smoke and lipopolysaccharide in a mouse model

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## ABSTRACT

*Ethnopharmacological relevance:* Hwangryunhaedok-tang is an oriental herbal formula treated to cure inflammation and gastric disorders in China, Japan, and Korea. We explored the protective effects of Hwangryunhaedok-tang water extract (HRWE) against airway pathophysiological changes caused by cigarette smoke (CS) and lipopolysaccharide (LPS) in a mouse.

*Materials and methods:* We performed quantitative analyses of five marker components, namely geniposide, baicalin, coptisine, plamatine, and berberine, using high-performance liquid chromatography. Animals were received CS exposure (1 h per day) for 7 days. LPS was administered intranasally on day 4. Mice were received HRWE at dose of 100 or 200 mg/kg for 1 h before CS exposure.

*Results:* Treatment with HRWE significantly suppressed the increased inflammatory cell count induced by CS and LPS exposure. In addition, reduction in IL-6, TNF-α and IL-1β in broncho-alveolar lavage fluid (BALF) was observed after HRWE treatment. HRWE not only decreased inflammatory cell infiltration in lung, but also decreased the expression of iNOS, NF-κB and matrix metallopeptidase (MMP)-9 in lung tissues.

*Conclusion:* This study showed that HRWE can attenuate respiratory inflammation caused by CS and LPS exposure. Therefore, HRWE has potential for treating airway inflammatory disease.

## 1. Introduction

Cigarette smoke (CS) consists of a complex mixture of more than 7000 chemical compounds and oxidants. It is a crucial inflammatory cause in the progression of chronic obstructive pulmonary disease (COPD) (Wahl et al., 2016; Nesi et al., 2016). COPD features recruitment of inflammatory cells into lung tissues and increased production of pro-inflammatory cytokines and proteases (Roh et al., 2010; Bucher et al., 2016). Since these inflammatory mediators exacerbate COPD, suppressing inflammatory responses is considered an important strategy for controlling COPD (Vaidyanathan and Damodar, 2015).

NF-κB transcription factor is considered as a key player in modulation of gene expression related with physiological responses induced by various injuries (Roh et al., 2010). In inflammatory responses, NF-κB is associated with induction of inflammatory mediator such as matrix metallopeptidase (MMP)-9, inducible-nitric oxide synthase (iNOS), interleukin (IL)-1β and tumor necrosis factor receptor (TNF)-α (Cha et al., 2016). These inductions ultimately produce nitric oxide, destroy normal alveolar structure, and cause infiltration of inflammatory cells into damaging lesions (O'Sullivan et al., 2014; Pang et al., 2015). In particular, pro-inflammatory mediators have been found in sputum and lavage samples of COPD patients (Mercer et al., 2005).

Herbal-based therapies have become increasingly popular among patients and physicians owing to their effectiveness and fewer side effects (Lee et al., 2014; Shin et al., 2011). Hwangryunhaedok-tang (orengedokuto in Japanese and huang-lian-jie-du-tang in Chinese) is an oriental herbal formula consisting of 4 herbs (*Gardeniae Fructus*, *Coptidis Rhizoma*, *Phellodendri Cortex*, and *Scutellariae Radix*). It has been used in China, Japan, and Korea to treat fever, inflammation,

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gastritis, and hypertension (Kwak et al., 2013; Kim et al., 2013; Seo et al., 2015). Recent studies have demonstrated that Hwangryunhaedok-tang possesses various pharmacological properties such as anti-inflammatory, antioxidant, anti-atherosclerotic, and anti-proliferative effects (Kwak et al., 2013; Seo et al., 2015). However, thus far, no study has investigated its protective effect against neutrophilic airway inflammation induced by CS.

The goal of experiment was to explore protective effects of Hwangryunhaedok-tang water extract (HRWE) against airway inflammatory responses caused by CS and lipopolysaccharide (LPS) exposure. To determine the possible mechanism of HRWE, the expression and production levels of inflammatory mediators were examined in CS and LPS-exposed mice.

## 2. Materials and methods

### 2.1. Plant materials

*Scutellaria baicalensis* Georgi (*Scutellariae Radix*, Labiatae), *Phellodendron amurense* Ruprecht (Phellodendri Cortex, Rutaceae) and *Coptis japonica* Makino (*Coptidis Rhizoma*, Ranunculaceae) were purchased from HMAX (Jecheon, Korea). *Gardenia jasminoides* Ellis (*Gardeniae Fructus*, Rubiaceae) was purchased from Omniherb (Yeongcheon, Korea). The 4 herbs were obtained from the Korea Institute of Oriental Medicine (2008–KE20–1, 2008–KE20–2, 2008–KE20–3, and 2008–KE20–4).

### 2.2. Chemicals and materials

Five reference standards (geniposide, baicalin, coptisine, palmatine, and berberine) with purity ≥98.0% were obtained from Wako (Osaka, Japan). High-performance liquid chromatography (HPLC)-grade water, methanol and acetonitrile were obtained from commercial company (J.T. Baker, NJ, USA). Phosphoric acid and sodium dodecyl sulfate (SDS) were obtained from commercial company (Daejung Chemicals & Metals, Daejeon, Korea and MP Biomedicals, OH, USA).

### 2.3. Preparation of Hwangryunhaedok-tang decoction

Hwangryunhaedok-tang decoction was prepared at KIOM. Briefly, four raw medicinal herbs (*Scutellariae Radix*, 2.5 kg; *Phellodendri Cortex*, 2.5 kg; *Coptidis Rhizoma*, 2.5 kg; and *Gardeniae Fructus*, 2.5 kg) were mixed and extracted in 100 L of distilled water using an electric extractor (Kyungseo Machine, Incheon, Korea). The solution was filtered through a standard sieve and freeze-dried using a vacuum controlled freeze dryer (PVT100, IlShinBioBase, Yangju, Korea). The amount of lyophilized Hwangryunhaedok-tang powder obtained was 1713.6 g (yield: 17.1%). Water was added to the powder to prepare the decoction.

### 2.4. HPLC analysis of Hwangryunhaedok-tang decoction

Simultaneous determination of the five markers in Hwangryunhaedok-tang decoction was conducted using a Shimadzu Prominence LC-20A system (Kyoto, Japan) equipped with DGU-20A3 online degasser, LC-20AT pumps, CTO-20A column oven, SIL-20AC auto sample injector, and SPD-M20A photodiode array (PDA) detector. Data were recorded and processed with LCsolution software. To separation of five marker components, we used a Phenomenex Gemini C18 column (250×4.6 mm, 5 µm, Torrance, CA, USA). The column temperature was maintained at 35 °C. The mobile phase consisted of 10% (v/v) acetonitrile, 0.2% SDS, phosphoric acid 200 µL/L (A), and acetonitrile (B). The gradient flow was set as follows: 10–40% B (0–20 min), 40–50% B (20–40 min), 50–100% B (40–50 min), 100–10% B (50–55 min), and 10% B (55–70 min). The flow rate and injection volume were 1.0 mL/min and 10 µL, respectively. For quantitative analysis, lyophilized Hwangryunhaedok-tang extract (250 mg) was dissolved in 25 mL of distilled water. Water extraction was performed at room temperature for 20 min by using an ultrasonicator (Branson 8510; Branson Ultrasonics, Danbury, CT, USA). Quantitative analysis of baicalin was performed after a 10-fold dilution of the sample solution. Each solution was filtered using a 0.2-µm membrane filter before injecting into HPLC system.

### 2.5. Animals

C57BL/6N male mice (Specific-pathogen-free, 6 weeks old and 20–25 g) were obtained from Samtako (Osan, Korea). They were divided into five groups under standard conditions with water and food provided *ad libitum*. All experimental protocols were granted by the Institutional Animal Care and Use Committee of Chonnam National University.

### 2.6. Establishment of the CS and LPS-caused airway inflammation mouse

To produced CS, we used 3R4F research cigarette (Kentucky reference cigarettes; Center for Tobacco Reference Products, University of Kentucky, USA). CS exposure was performed according to previously described (Shin et al., 2014). The animals were received CS exposure for 1 h per day in exposure chamber. The exposure periods is 7 days. LPS (10 µg per mouse) was exposed to animals by intranasal instillation under anesthesia on day 4. HRWE was treated to animals at doses of 100 or 200 mg/kg by oral gavage 1 h before CS exposure for 7 days.

### 2.7. Collection of broncho alveolar lavage fluid (BALF)

The animals were anesthetized by Zoletil 50 (25 mg/kg; Virbac Korea, Seoul, Korea) at 48 h after LPS instillation,. A tracheostomy was conducted as previously study (Shin et al., 2014). The cells were resuspended in PBS after collecting the supernatant. The suspended cells were spread onto glass microscope slides; each slide was air-dried and then stained with Diff-Quik® reagent (IMEB Inc., San Marcos, CA, USA).

### 2.8. Measurement of proinflammatory mediators in BALF

The TNF-α, IL-1β and IL-6 were analyzed by ELISA kits (R & D System, CA, USA) according to instructions of manufacturer. The absorbance (450 nm) was measured using a microplate reader (Bio-Rad, CA, USA).

### 2.9. Immunoblotting

Immunoblot was performed as previously study (Shin et al., 2014). The primary antibodies were anti-pNF-κB (Abcam, MA, USA), anti-NF-κB (Abcam), anti-β-actin (Cell Signaling, MA, USA) and anti-iNOS (Santa Cruz, MA, USA).

### 2.10. Gelatin zymography

Gelatin zymography was conducted as previously studies (Shin et al., 2014) to determine gelatinase activity. The white band on a gelatin gel was represented as proteolytic areas induced by MMP-9.

### 2.11. Lung tissue histopathology and immunohistochemistry

The lungs from each animal were collected, immersed, embedded in paraffin and sliced into 4-µm thick, which were then stained with hematoxylin and eosin solution (Sigma-Aldrich, CO, USA) and MMP-9 antibody (Abcam).

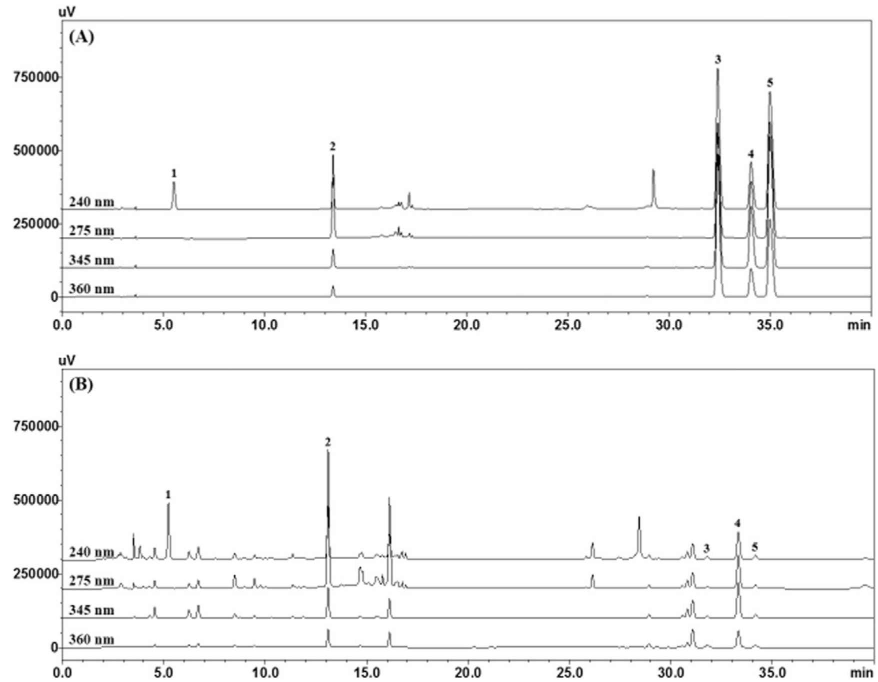


**Fig. 1.** HPLC chromatograms of standard mixtures (A) and HRWE decoctions (B) at different detection wavelengths. Geniposide (**1**), baicalin (**2**), coptisine (**3**), palmatine (**4**), and berberine (**5**).

### 2.12. Statistical analysis

Data were presented as means ± standard deviation (SD). Statistical analysis was adjust to ANOVA followed by the Dunnett's test. Significance was considered as P-values of less than 0.05.

## 3. Results

### 3.1. HPLC analysis of HRWE

As shown in Fig. 1, geniposide, baicalin, coptisine, palmatine, and berberine were separated within 40 min. Geniposide, baicalin, coptisine, palmatine, and berberine were eluted at approximately 5.22, 13.10, 31.88, 33.45, and 34.27 min retention time, respectively. The amounts of geniposide, baicalin, coptisine, palmatine, and berberine detected were found to be 36.61, 30.53, 0.94, 10.28, and 1.32 mg/g, respectively.

### 3.2. HRWE suppresses the increased inflammatory cell count caused by CS and LPS exposure

To ascertain the effect of HRWE on inflammatory cells in BALF, cytological classification and cell counts were performed. As shown in Fig. 2, CS and LPS exposure induced an increase in inflammatory cell count particularly neutrophil numbers compared to that in vehicle control mice. Roflumilast-treated animals had meaningfully decreased neutrophil count compared to the CS and LPS-exposed animals. Neutrophil count in mice was dose-dependently decreased by HRWE treatment compared with CS and LPS-exposed mice.

### 3.3. HRWE inhibits pro-inflammatory cytokines caused by CS and LPS

As shown in Fig. 3, CS and LPS-exposed animals exhibited a marked elevation of IL-6 and TNF-α compared to vehicle control mice. However, HRWE-treated animals displayed significant decreases in IL-6, IL-1β and TNF-α in BALF in comparison to CS and LPS-exposed animals.

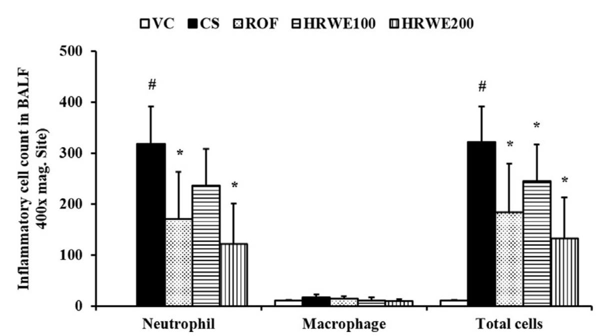


**Fig. 2.** HRWE decreased inflammatory cell counts in BALF. VC: vehicle control animal; CS: CS and LPS-exposed animal; ROF: roflumilast (10 mg/kg)+CS and LPS-exposed animal; HRWE100: HRWE (100 mg/kg)+CS and LPS-exposed animal; HRWE200: HRWE (200 mg/kg)+CS and LPS-exposed animal. Values are presented as means ± SD. #, $P < 0.05$ compared with vehicle control animals; *, $P < 0.05$ compared with CS animals.

### 3.4. HRWE suppresses expression of pro-inflammatory proteins in lung tissues caused by CS and LPS exposure

To assess effect of HRWE on iNOS and NF-κB in CS and LPS-exposed animals, we evaluated protein expression by western blotting. CS and LPS-exposed animals exhibited a meaningful increase in iNOS expression in comparison to vehicle control mice (Fig. 4A). However, HRWE-treated animals showed an obvious reduction in iNOS expression in comparison to CS and LPS-exposed mice. Similarly, the NF-κB phosphorylation was obviously elevated in CS and LPS-exposed animals compared to vehicle control mice (Fig. 4B). By contrast, HRWE-treated animals displayed an obvious reduction in phosphorylation of NF-κB in comparison to CS and LPS-exposed mice.

### 3.5. HRWE decreases inflammatory cell infiltration caused by CS and LPS exposure

To analyze the effect of HRWE on histopathological changes caused by CS and LPS exposure, we performed H & E staining of lung tissue. As shown in Fig. 5, CS and LPS-exposed animals displayed obvious

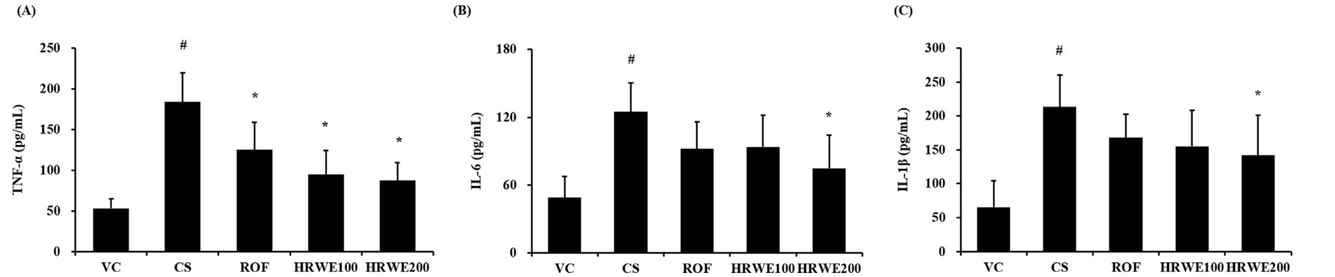


**Fig. 3.** HRWE decreased pro-inflammatory cytokine caused by CS and LPS exposure. (A) TNF-α, (B) IL-6, and (C) IL-1β. VC: vehicle control animals; CS: CS and LPS-exposed animals; ROF: roflumilast (10 mg/kg)+CS and LPS-exposed animals; HRWE100: HRWE (100 mg/kg)+CS and LPS-exposed animals; HRWE200: HRWE (200 mg/kg)+CS and LPS-exposed animals. Values are presented as means ± SD. #, $P < 0.05$ compared with vehicle control animals; *, $P < 0.05$ compared with CS animals.

recruitment of inflammatory cells in peribronchiolar and alveolar lesions. In contrast, HRWE-treated animals had decreased inflammatory cell infiltration compared to CS and LPS-exposed mice.

### *3.6. HRWE reduces MMP-9's expression and activity induced by CS and LPS exposure*

In Fig. 6A, MMP-9 expression in lung of CS and LPS-exposed animals was increased in comparison to vehicle control mice. In contrast, HRWE-treated animals had reduced MMP-9 expression in lung in comparison to CS and LPS-exposed mice. In the zymography assay (Fig. 6B), CS and LPS-exposed animals displayed a marked elevation of MMP-9 activity compared to vehicle control mice, whereas HRWE-treated animals displayed a significant decrease compared to CS and LPS-exposed animals.

## 4. Discussion

We explored the ameliorative effect of HRWE against neutrophilic airway inflammatory alterations caused by CS and LPS exposure. HRWE significantly reduced inflammatory cell count in BALF. Furthermore, HRWE-treated animals had meaningfully lower IL-1β, IL-6 and TNF-α than CS and LPS-exposed animals. These results were observed in histopathological analysis of lung tissues. Finally, HRWE effectively suppressed the iNOS, MMP-9, and NF-κB expression caused by CS and LPS exposure.

COPD is featured as excessive mucus secretion and limited airflow. CS is believed to be the main risk factor in progress of COPD. It contributes to irreversible pathological changes in the lung (Dong et al., 2015a, 2015b; Nesi et al., 2016). CS exposure induces an elevation of inflammatory cell counts, leading to constant airway remodeling. It also promotes production of pro-inflammatory cytokines and protease, thus aggravating COPD (Nakamura et al., 2015; Jung et al., 2016). Moreover, pro-inflammatory cytokines are associated with exacerbation of the inflammatory response in COPD development (Ge et al., 2016). Recently, studies have demonstrated that IL-1β, IL-6 and TNF-α are markedly increased in cigarette smoke extract induced human macrophage cells and CS exposed rats (Mortaz et al., 2015; Wang et al., 2015). Our results showed that HRWE treatment significantly suppressed increases in inflammatory cell count and pro-inflammatory cytokines induced by CS and LPS exposure. These results indicate that HRWE can decrease inflammatory cells and pro-inflammatory cytokines induced by CS exposure.

MMPs, a one of endopeptidases, can destroy extracellular matrix proteins. They participate in a variety of physiological and pathological processes (Snitker et al., 2013). In the development of COPD, MMPs are involved in various processes, including production of growth factors and cytokines as well as inflammatory-cell migration. MMP-9 is the largest and the most structurally complex member. It can destroy normal alveolar structure, aggravate inflammatory response in lung

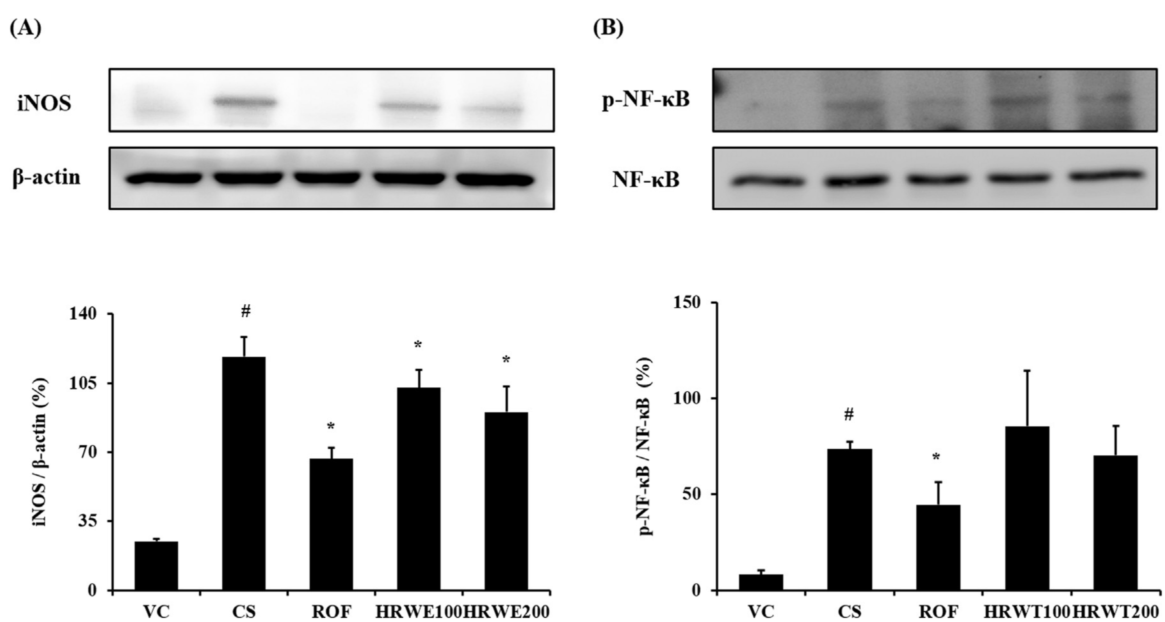


**Fig. 4.** HRWE reduced expression of pro-inflammatory proteins in lung tissues caused by CS and LPS. (A) Representative figure of protein expression, and (B) Quantitative analysis of protein expression. VC: vehicle control animals; CS: CS and LPS-exposed animals; ROF: roflumilast (10 mg/kg)+CS and LPS-exposed animals; HRWE100: HRWE (100 mg/kg)+CS and LPS-exposed animals; HRWE200: HRWE (200 mg/kg)+CS and LPS-exposed animals. Values are presented as means ± SD. #, $P < 0.05$ compared with vehicle control animals; *, $P < 0.05$ compared with CS animals.

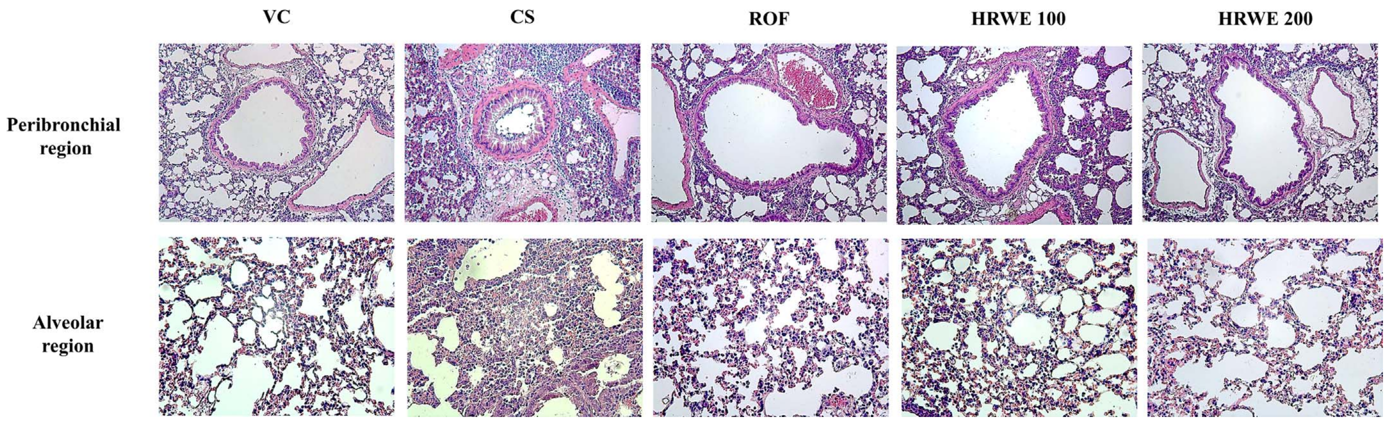


**Fig. 5.** HRWE inhibited inflammatory cell infiltration caused by CS and LPS. Inflammatory infiltration in the peri-bronchial region and alveolar region was revealed by H & E staining. VC: vehicle control animals; CS: CS and LPS-exposed mice; ROF: roflumilast (10 mg/kg)+CS and LPS-exposed mice; HRWE100: HRWE (100 mg/kg)+CS and LPS-exposed mice; HRWE200: HRWE (200 mg/kg)+CS and LPS-exposed mice.

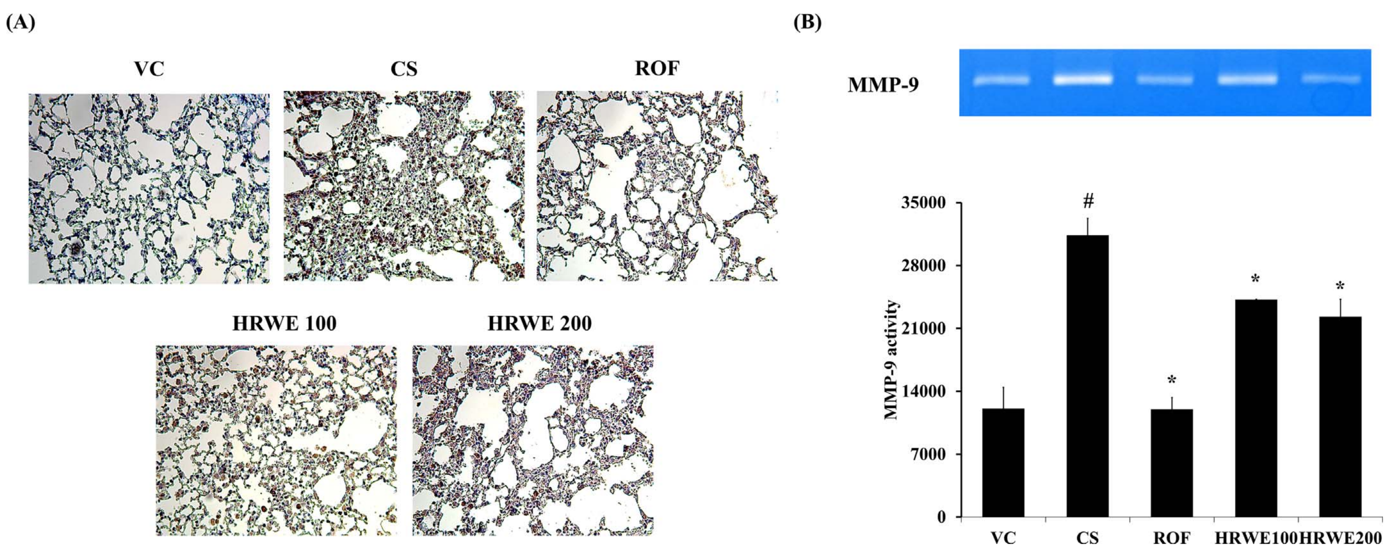


**Fig. 6.** HRWE suppressed the expression and activity of MMP-9 in the lung tissue induced by CS and LPS exposure. VC: vehicle control mice; CS: CS and LPS exposed mice; ROF: roflumilast (10 mg/kg)+CS and LPS exposed mice; HRWE100: HRWE (100 mg/kg)+CS and LPS exposed mice; HRWE200: HRWE (200 mg/kg)+CS and LPS exposed mice. #Significantly different from the vehicle control group, $P < 0.05$; *Significantly different from the CS group, $P < 0.05$.

tissues, and induce emphysema, resulting in loss of lung function (Shin et al., 2014). In clinical trials, MMP-9 protein was observed in the sputum and lavage samples of COPD patients (Pang et al., 2015; Cane et al., 2016). In our study, HRWE treatment effectively suppressed elevated expression and activity of MMP-9. These results suggest that the ameliorative effect of HRWE might be associated with its inhibition of MMP-9 in animals with CS and LPS exposure.

NF-κB is central in the modulation of MMP-9 expression during inflammatory responses (Lee et al., 2009; Edwards et al., 2009). NF-κB activation is induced by pro-inflammatory stimuli such as LPS and TNF-α, thus causing the degradation of IkB (Shin et al., 2013). It is then translocated to the nucleus. It can induce transcription of iNOS, COX2, and MMP-9. CS is a potential stimulus that can activate NF-κB (Schuliga, 2015). CS induces the phosphorylation of NF-κB and eventually produces chemokines, cytokines and MMPs. Indeed, NF-κB activation is markedly increased in COPD patients (Wang et al., 2015). Such evidence suggests that suppression of NF-κB might be an important step in controlling COPD (Edwards et al., 2009). CS and LPS-exposed mice had meaningfully elevated phosphorylation of NF-κB compared to vehicle control mice, as reported in previous studies in present study. However, HRWE treatment inhibited activation of NF-κB caused by CS and LPS exposure. These results indicate that HRWE can suppress inflammatory responses caused by CS and LPS exposure by downregulating NF-κB activation.

The protective effects of HRWE might be due to its physiologically active components, including geniposide, baicalin, coptisine, palmatine, and berberine. Previous studies have demonstrated that these components have a range of pharmacological properties. In particular, geniposide, baicalin, coptisine, and berberine have strong anti-inflammatory properties closely associated with the suppression of NF-κB activation (Shin et al., 2014; Dong et al., 2015a, 2015b; Fu et al., 2015; Wu et al., 2016). Moreover, it has been recently reported that Hwangryunhaedok-tang can inhibit NF-κB phosphorylation in LPS-stimulated human vascular endothelial cells (Wu et al., 2012).

In conclusion, our study displayed that HRWE could inhibit neutrophilic airway inflammation caused by CS and LPS exposure. This effect might be closely associated with reduction in NF-κB activation. This study suggest that HRWE has potential for treating airway inflammatory conditions such as COPD.

## Conflict of interest

We have no competing financial interests.

## Acknowledgments

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